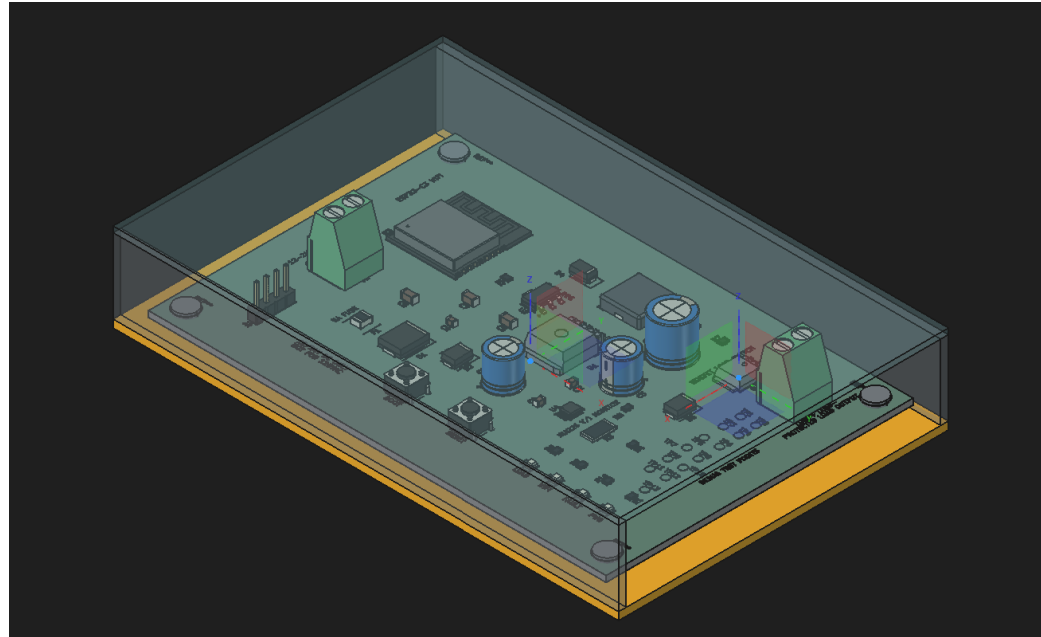


Wi-Fi SmartFuse PCB

12/24V DC Smart Relay, Fuse Protection, and Power Monitoring Module

Personal Engineering Portfolio Project - Naman Arora



Final FreeCAD enclosure mockup: PCB mounted inside a transparent protective housing.

PCB Design

Completed in KiCad

3D Mockup

Completed in FreeCAD

Current Stage

Virtual prototype; not manufactured yet

1. Project idea in simple words

The problem In many DC power systems, a simple fuse or relay can disconnect a load, but it does not tell the user what is happening. For motors, fans, LED loads, battery systems, and automation panels, it is useful to know voltage, current, power, and fault status. Working with low and medium voltage systems and high-power motor management relays helped me understand why protection and monitoring should work together.	The solution This project is a compact 12/24V DC smart fuse and relay board. It uses an ESP32-C3 for Wi-Fi control, an INA226 for voltage/current monitoring, a MOSFET for DC load switching, and protection parts at the input/output. The goal is to make DC load control safer, easier to observe, and easier to debug.
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2. How it works

12/24V DC input	Fuse + TVS protection	5V + 3.3V regulation	ESP32-C3 control	INA226 measurement	MOSFET load output
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3. Possible market picture

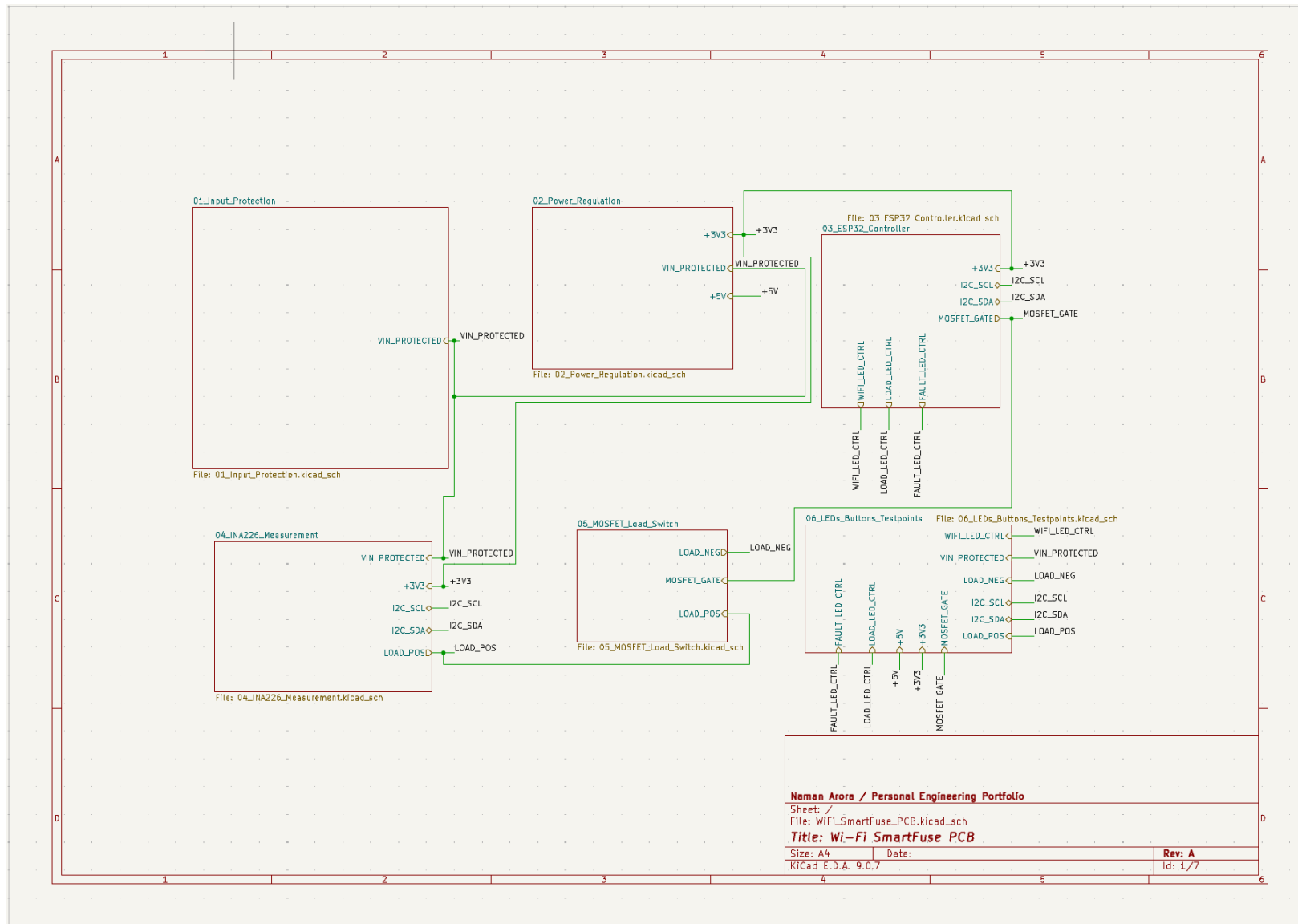
After real testing and proper component selection, this type of board could fit into low and medium voltage DC power products where users need switching, protection, and monitoring in one small module.

- DC motor and fan control modules
- Battery, solar, and 12/24V automation panels
- Robotics, small vehicles, and lab power systems
- Smart load monitoring for maintenance and fault detection
- Educational or evaluation boards for DC protection and relay control

Safety note: This design is for low-voltage DC work only. It is not for 120 VAC or mains power.

4. Complete schematic structure

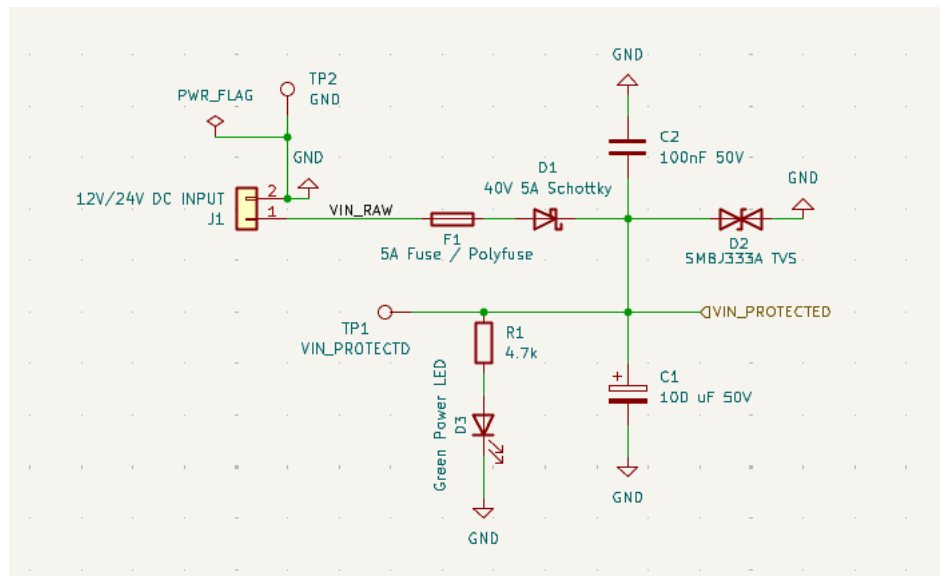
The design was divided into small sheets so every function could be checked separately: input protection, power regulation, ESP32 controller, current measurement, MOSFET load switch, and LEDs/test points.



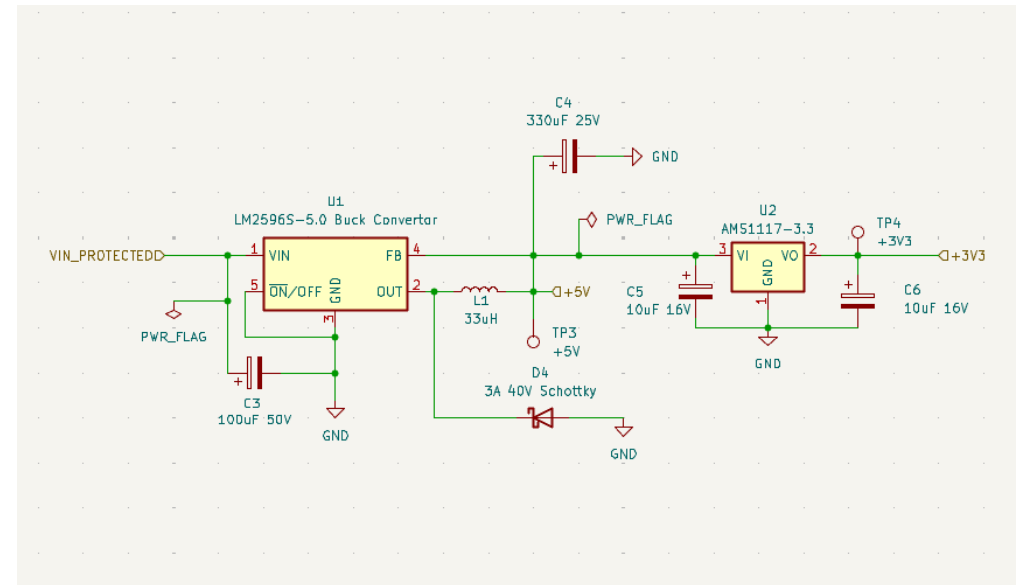
Top-level KiCad schematic showing how the main blocks connect together.

5. Power path and regulation

The input side protects the board from basic DC power problems. The regulator section converts the protected input into 5V and 3.3V rails for the ESP32 and sensors.



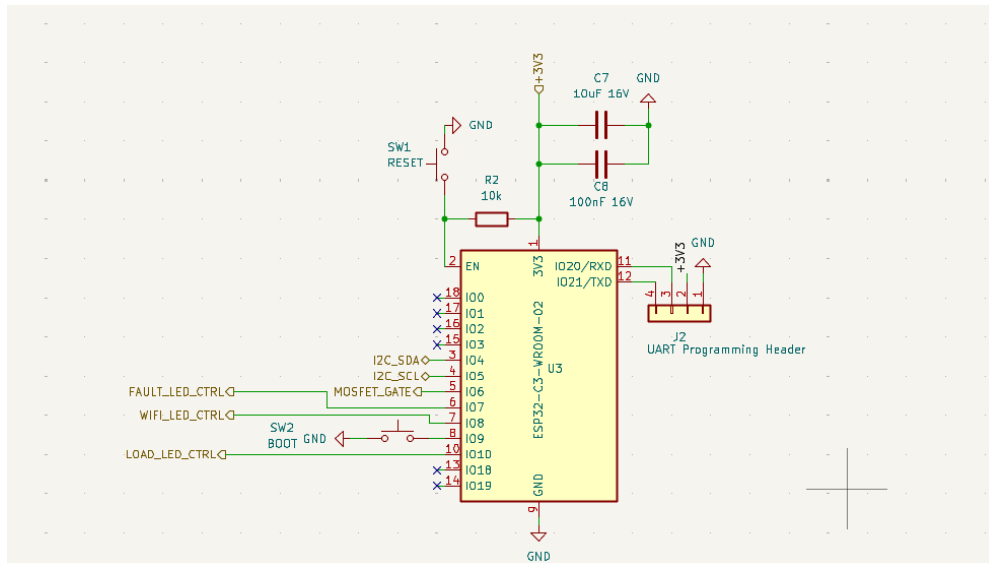
Input protection: DC input terminal, fuse, Schottky diode, TVS diode, capacitors, and test points.



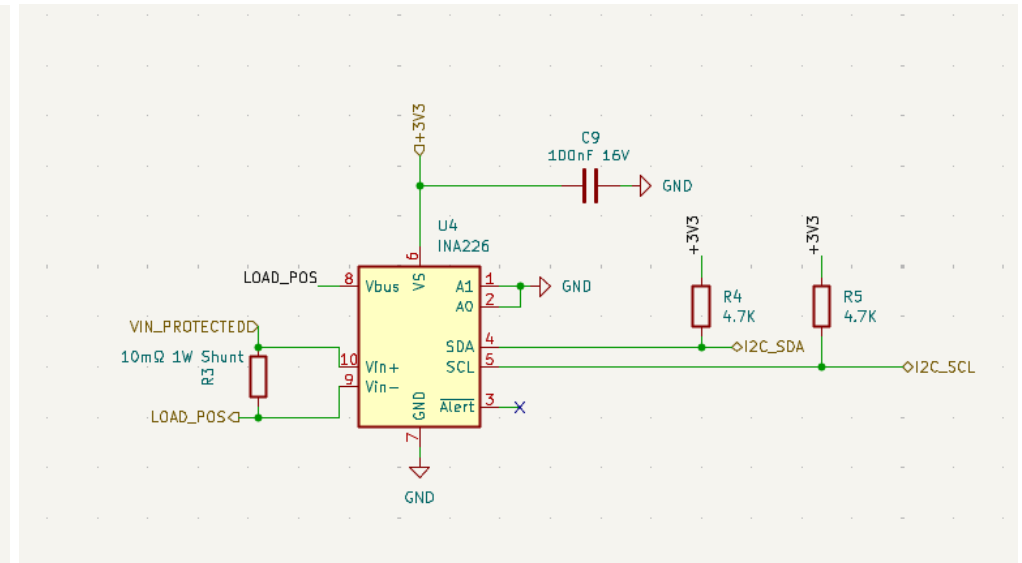
Power regulation: LM2596 buck converter creates 5V, then AMS1117 creates 3.3V.

6. Control and measurement

The ESP32-C3 controls the MOSFET gate, status LEDs, and buttons. The INA226 measures voltage and current so the firmware can detect overload or fault conditions.



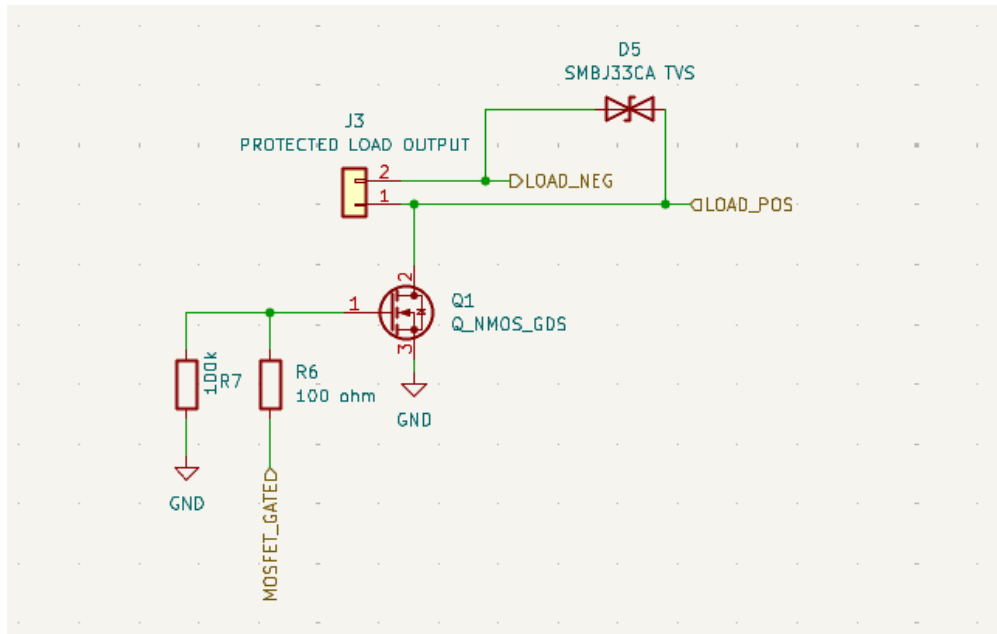
ESP32-C3 controller section with reset, boot button, UART programming header, I2C, and control signals.



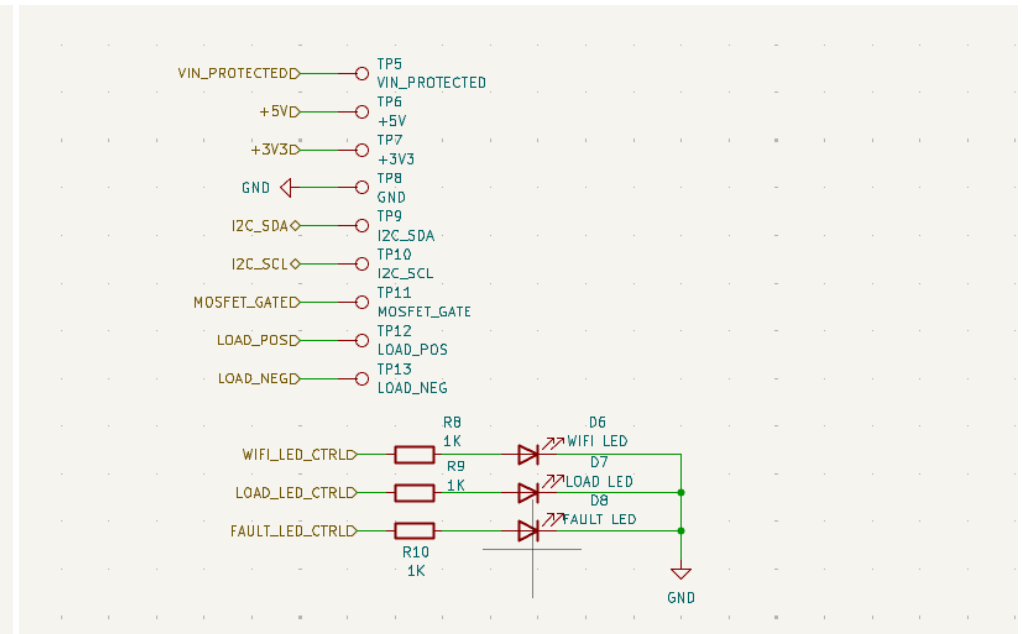
INA226 measurement section for bus voltage and current sensing through the shunt resistor.

7. Load switching, LEDs, and debug points

The MOSFET section switches the DC load. The LED and test point section makes the board easier to test, debug, and explain during a demo.



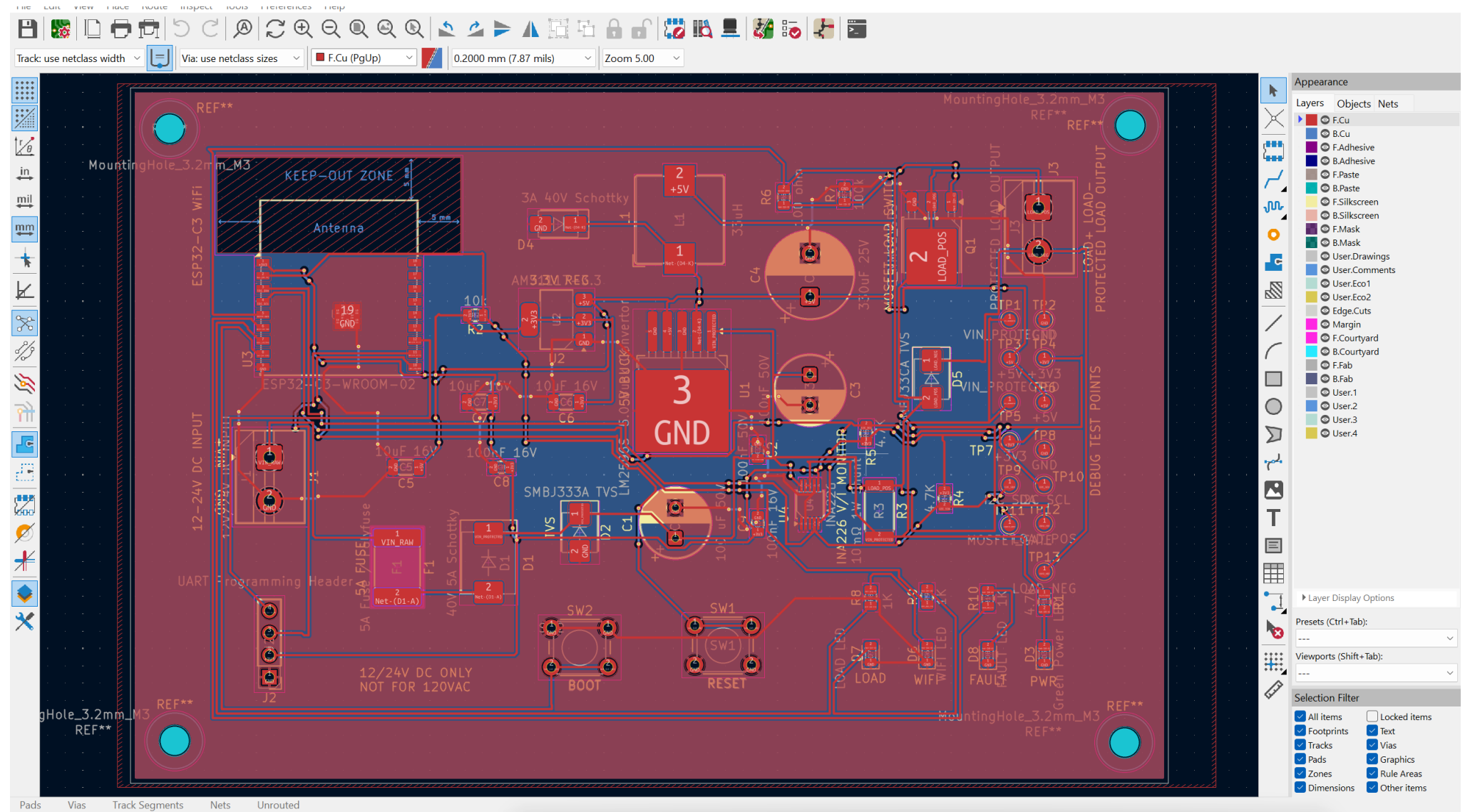
MOSFET low-side load switch with gate resistor, pulldown, and output protection.



LEDs, buttons, and debug test points for power, load, Wi-Fi, fault, I2C, and output nets.

8. PCB layout

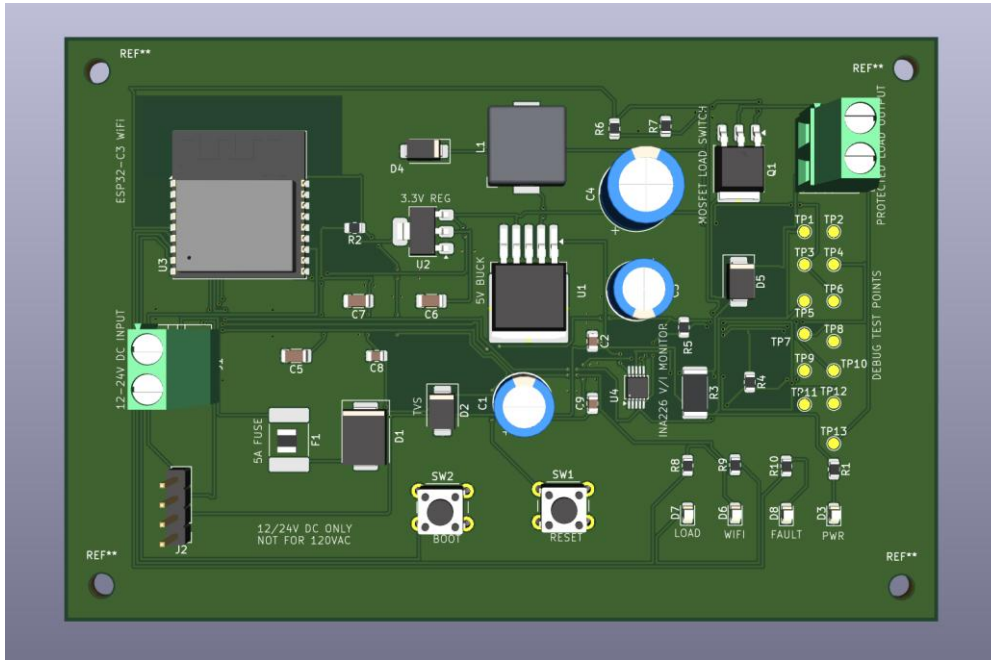
The PCB layout was completed with clear input/output areas, test points, mounting holes, safety labels, and a keep-out region around the ESP32 antenna.



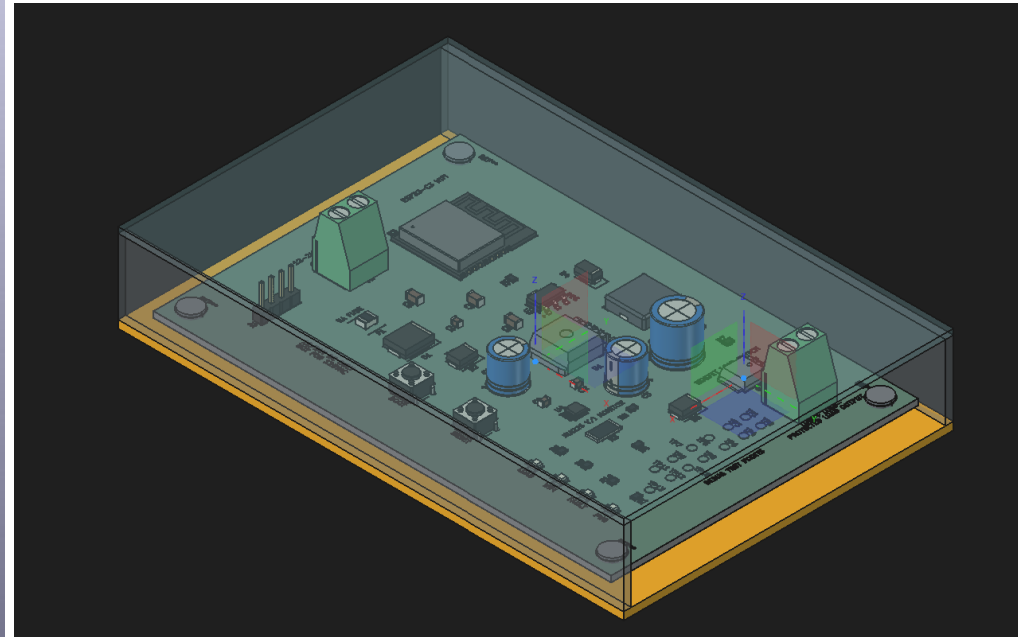
KiCad PCB layout with copper pours, components, labels, mounting holes, and antenna keep-out zone.

9. 3D prototype and enclosure mockup

The board was exported into 3D so the mechanical design could be checked before manufacturing. The FreeCAD mockup shows how the PCB can sit on standoffs inside a simple protective housing.



KiCad 3D render of the assembled PCB with realistic parts and labels.



FreeCAD enclosure mockup with base plate, standoffs, screw heads, front/back walls, and transparent cover.

10. Next steps

- Finalize the real bill of materials with exact part numbers and current ratings.
- Load the ESP32 firmware and test first with a current-limited 12V bench supply.
- Validate INA226 readings, MOSFET switching, LED behavior, and overcurrent trip logic.
- Run thermal/current testing before any manufacturing or real-world use.
- Later, export Gerbers, BOM, and pick-and-place files for PCBA manufacturing.

Project status: design and virtual enclosure mockup completed. Physical manufacturing is a future step.